



MCCA
GLOBAL TEC
FORUM

Autonomous Vehicles:

Navigating the legal and regulatory issues
of a driverless world

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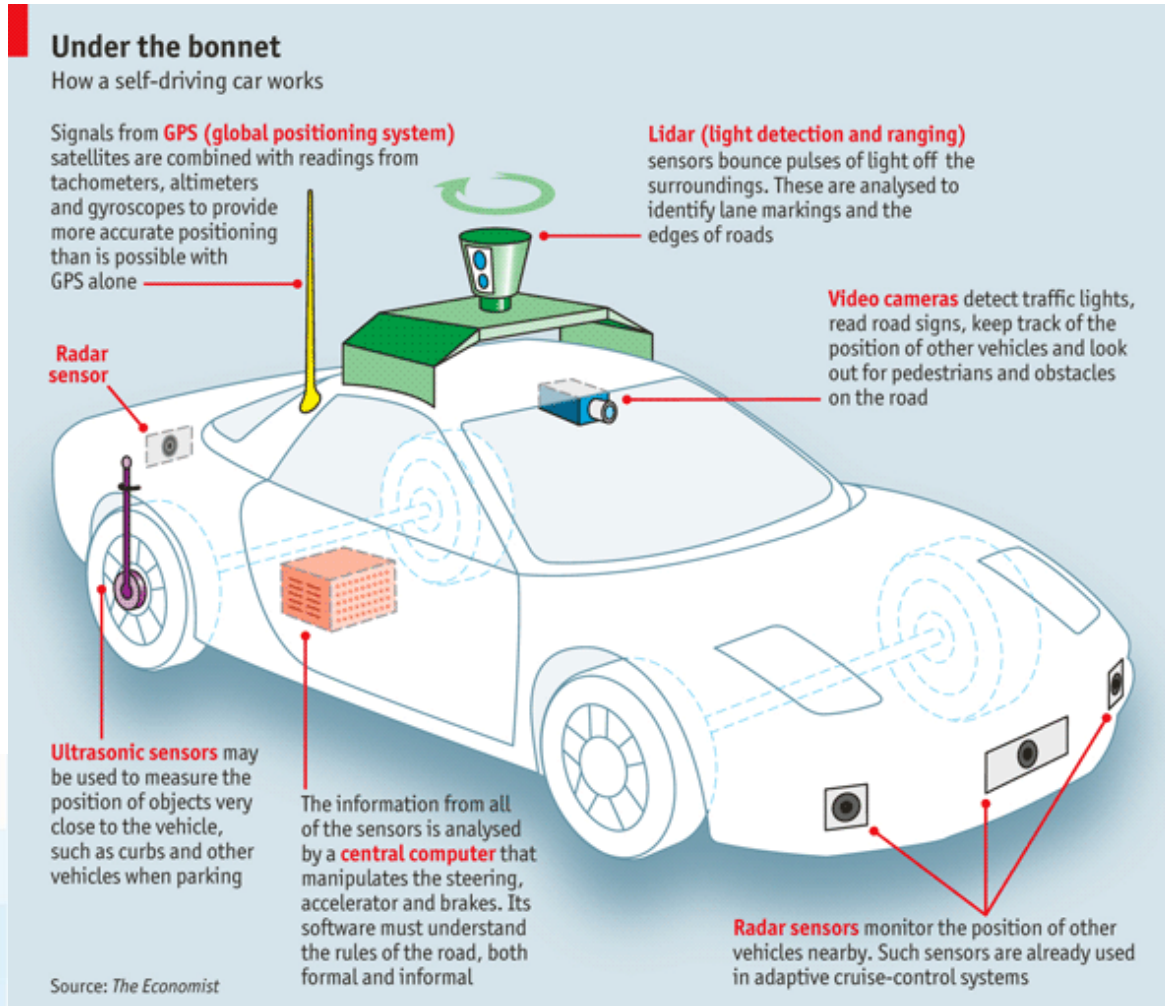
- Keolis – AVs in Public Transportation



A high-angle photograph of a white laptop and a white mouse on a dark wooden desk. A white modern chair is visible in the lower-left foreground. A semi-transparent grey rectangle with a thin blue border is centered over the laptop, containing the word "Introduction" in bold black text.

Introduction

Basic Physical Ecosystem of an Autonomous Vehicle



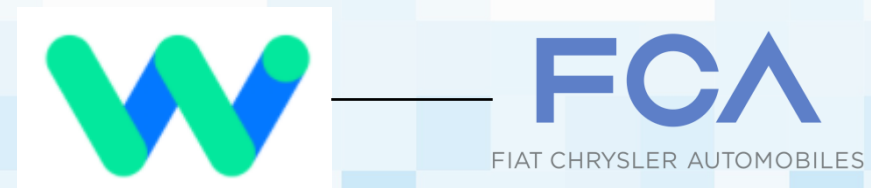
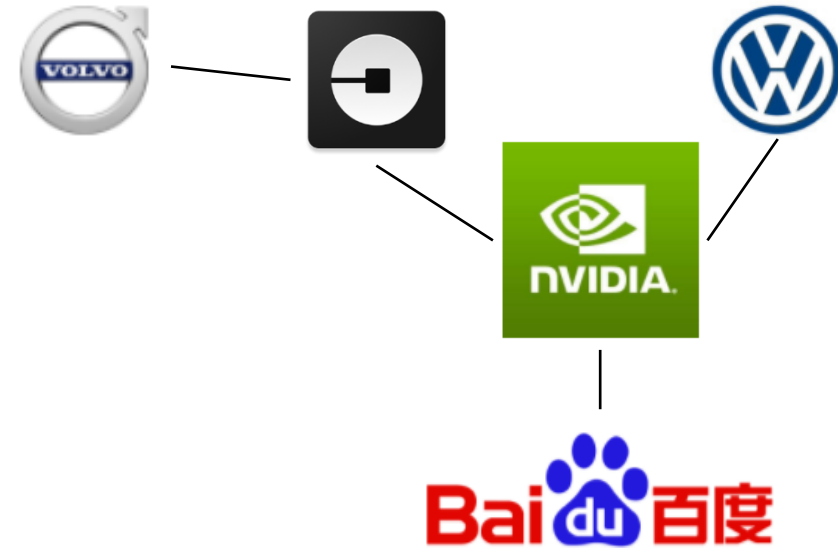
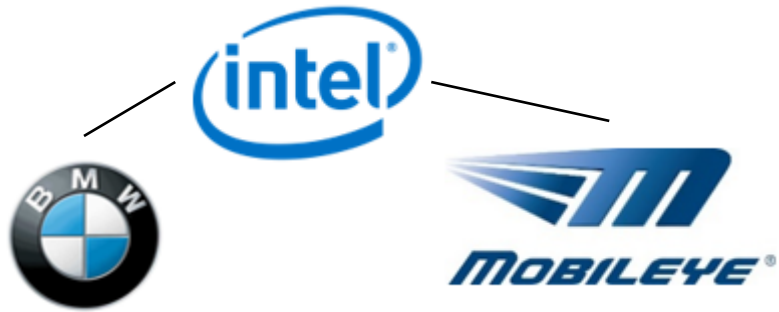
- Global Positioning System (GPS)
- Light Detection and Ranging (LIDAR)
- Cameras (Video)
- Ultrasonic Sensors
- Central Computer
- Radar Sensors
- Dedicated Short-Range Communications-Based Receiver (not pictured)

Source: *The Economist*, “How does a self-driving car work?”

Key Physical Components of Autonomous Vehicles

- **Cameras** – Provide real-time obstacle detection to facilitate lane departure and track roadway information (like road signs).
- **Radar** – Radio waves detect short & long-range depth.
- **LIDAR** – Measures distance by illuminating target with pulsed laser light and measuring reflected pulses with sensors to create 3-D map of area.
- **GPS** – Triangulates position of car using satellites. Current GPS technology is limited to a certain distance. Advanced GPS is in development.
- **Ultrasonic Sensors** – Uses high-frequency sound waves and bounce-back to calculate distance. Best in close range.
- **Central Computer** – “Brain” of the vehicle. Receives information from various components and helps direct vehicle overall.
- **DRSC - Based Receiver** – Communications device permitting vehicle to communicate with other vehicles (V2V) using DSRC, a wireless communication standard that enables reliable data transmission in active safety applications. NHTSA has promoted the use of DSRC.

Strategic Partnerships



A high-angle photograph of a workspace. A silver laptop is open on a dark wooden desk. To its right is a white computer mouse. In the foreground, the back of a white, modern-style chair is visible. A semi-transparent grey rectangle with a thin blue border is centered over the laptop, containing the text "Federal and State Regulations".

Federal and State Regulations

Federal Regulation of Autonomous Vehicles

- DOT released new guidance on autonomous vehicles in September 2017, titled “Automated Driving Systems (ADS) 2.0: A Vision for Safety”. Replaces guidance issued in 2016 by Obama DOT.
 - Guidance identifies 12 safety elements: (1) system safety; (2) operational design domain; (3) object and event detection and response; (4) fallback (minimal risk condition); (5) validation methods; (6) human machine interface; (7) vehicle cybersecurity; (8) crashworthiness; (9) post-crash ADS behavior; (10) data recording; (11) consumer education and training; and (12) federal state and local laws.
 - Recommends that entities involved in ADS testing and deployment demonstrate how they address the 12 safety elements by publishing a Voluntary Safety Self-Assessment.
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A high-angle photograph of a modern workspace. A silver laptop is open on a dark wooden desk. To its right is a white computer mouse. In the foreground, the back of a white, curved office chair is visible. A semi-transparent grey rectangle with a thin blue border is centered over the laptop and mouse, containing the text "Product Liability".

Product Liability

Managing Liability Among Parties

- Warranties and indemnifications should clearly define scope, responsibility and liability
- Responsibility for maintenance, repairs and updates should be defined
- Liability between automaker, technology company and vehicle owner/operator should be defined
- Responsibility for compliance with federal, state and local laws and regulations should be defined



Common Security Vulnerabilities

- **Software Glitches** – Connected vehicles today contain more than 100 million lines of code. More code means more opportunity for bugs and mistakes. Glitches, even when inadvertent, can be exploited.
- **No Single Source of Knowledge of or Control Over Source Code** – Software for different components of connected vehicles is being written by different developers, installed by different suppliers, and no one source has knowledge of or control over the source code.
- **Increase Use of Apps Leave Vulnerabilities** – Consumers are using an increasing number of smartphone apps to interface with their connected cars and help run certain functions. Researchers have already demonstrated weaknesses in some of these apps. Likely to see spread in use of malware.
- **Need for Constant Updates May be Overlooked** – With the increased use of connected features comes an increased need for continuous updates to fix glitches and help protect vehicles. There is a risk these updates could be overlooked or that malicious actors could infect routine updates.

Key Cybersecurity Takeaways

- **Provide Multi-layered protection** – Beginning at level of individual ECUs, moving up a level to include software to protect vehicle's internal network by examining all network communications, and building in mechanisms to stop attacks from advancing within network.
 - **Defend against externally-facing potential gateways** – Ensure weakest links in car's security are viewed as potential threats and defenses are built into system. This is particularly true to infotainment or similar externally-facing mechanisms that are developed or utilized by multiple external entities.
 - **Ensure vendors and suppliers have strong security** – Connected and autonomous vehicles are made up of subparts and subsystems. It is critical to review and monitor vendor and supplier policies and practices.
 - **Promote timely updates** – Companies should push timely and effective fixes as soon as problems are identified.
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